Module 16: Capstone Systems Synthesis - Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Connect concepts from at least three modules in one explanation.
2. Trace a mechanism across biological scales.
3. Identify feedback loops, constraints, and tradeoffs.
4. Match evidence types to claims at different scales.
5. Revise a systems explanation using new evidence.

Topics

- Biology across scales
- Mechanism chains
- Feedback and constraint
- Evidence from multiple modules
- Capstone explanation and revision

Key Terms to Know

- **System** - Interacting parts whose relationships shape outcomes.
- **Scale** - Level of biological organization.
- **Mechanism** - Causal process that explains how something happens.
- **Feedback loop** - Circular causal relationship that amplifies or reduces change.
- **Constraint** - Limit that shapes what a system can do.
- **Tradeoff** - Benefit in one function paired with cost elsewhere.
- **Synthesis** - Combining ideas into a coherent explanation.
- **Evidence chain** - Linked evidence supporting each step of a claim.

Core Contents

1. Biological explanations become stronger when they connect molecules, cells, organisms, populations, and ecosystems.
2. Mechanism chains show how one change can produce effects across scales.
3. Feedback loops and constraints explain why biological systems do not change without limits.
4. Evidence from labs, models, and observations must match the scale of the claim.
5. Capstone reasoning combines concepts into a clear claim that can be tested and revised.

Study Tips

1. Start with the module's central claim before memorizing vocabulary.
2. Use the same-numbered lab as the evidence check for the module.
3. Explain one mechanism in words, then redraw it as a simple diagram.
4. Answer retrieval questions without notes, then revise with the terms list.

Connected Lab

- `lab-16_capstone-systems-synthesis.md`

Generated Visuals

- Module 16: Capstone Systems Synthesis Concept Map
- Module 16: Capstone Systems Synthesis Process Model
- Module 16: Capstone Systems Synthesis Retrieval Card